WEARABLE INCONTINENCE MONITORING SYSTEM

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PROBLEM STATEMENT

The Problem

Infants & Elderly

Moisture Exposure

Prolonged wetness causes severe skin irritation and infections in vulnerable patients

Delayed Awareness

Caregivers are not immediately aware when a diaper becomes wet, causing hygiene delays

Health Impact

Delayed response negatively affects both hygiene and the overall health of the patient

OBJECTIVES

01

Detect Moisture in Real Time

The system continuously monitors wetness using a moisture sensor and instantly captures changes in diaper condition

02

Process Data via Microcontroller

The Arduino Uno R3 reads the analog sensor signal, applies threshold logic, and determines whether to trigger an alert

03

Alert Caregivers Immediately

Visual (LED) and audio (buzzer) indicators are activated instantly, with status shown on LCD display for caregiver awareness

METHODOLOGY: System Architecture

1

Moisture Sensor

Detects wetness
Sends analog signal
0–1023 ADC range

2

Arduino Uno R3

Reads sensor data
Applies threshold logic
Triggers outputs

3

Output Devices

LED (visual)
Buzzer (audio)
LCD display (status)

Signal Flow: Sensor detects moisture → Arduino processes signal → LED / Buzzer / LCD activated

CIRCUIT DESIGN

Tinkercad Design Suite

Simulation Tool

"A potentiometer was used to simulate the moisture sensor behaviour during initial testing phases."

Arduino Uno R3

Central microcontroller — reads analog sensor values and controls all output devices

Potentiometer

Simulates moisture sensor input — used during tinkercad simulation and circuit testing

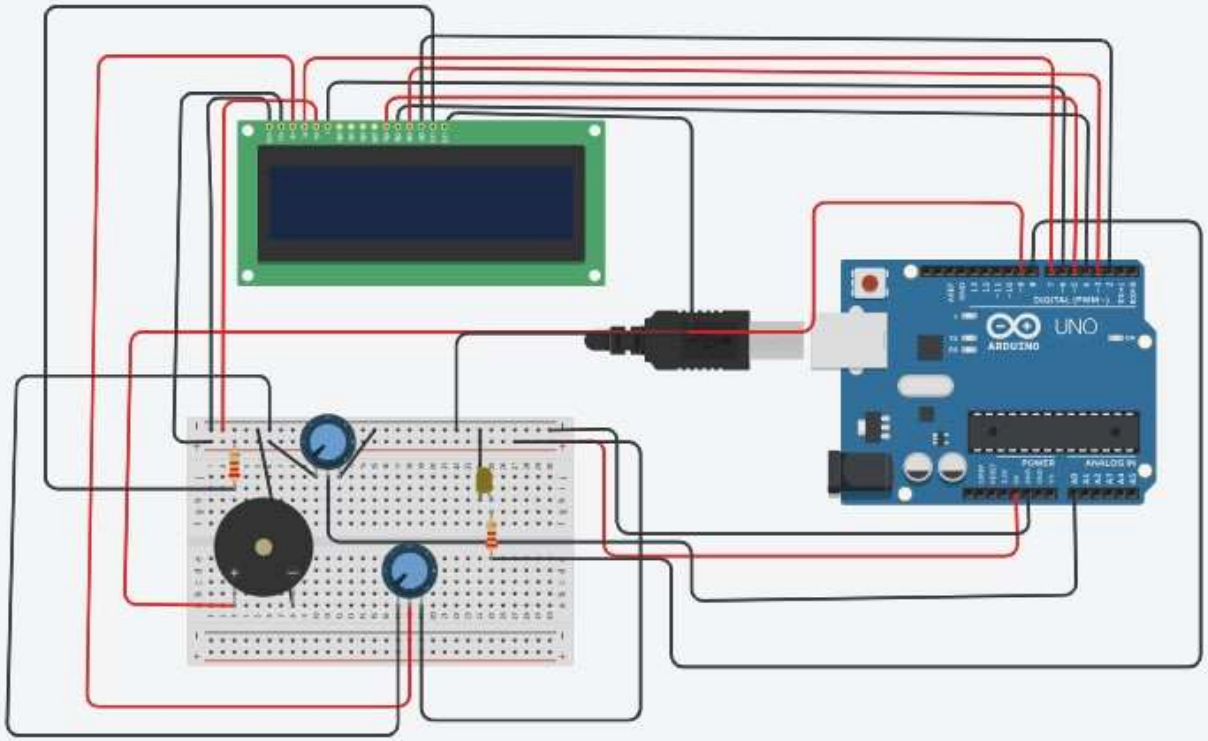
LED

Visual indicator — lights up to signal wet/dry status to the caregiver

Buzzer & LCD

Audio alert + text display — combined output for immediate and clear caregiver notification

CIRCUIT DESIGN



SENSOR EXPLANATION

01

Moisture Present

Water bridges the sensor electrodes — resistance drops significantly when wetness is detected

02

Analog Signal Rises

Decreased resistance → increased current flow → higher analog voltage value (0–1023 ADC range)

03

MCU Reads Signal

Arduino's ADC reads the analog pin value and compares it against the programmed moisture threshold

Key Principle: Moisture ↑ → Resistance ↓ → Current ↑ → Analog Value ↑ → Alert Triggered

CODE EXPLANATION

Arduino Logic Flow

01 Read analog value from sensor pin using ADC

02 Compare value against threshold (e.g. 500)

03 If value > threshold → WET state detected

04 Activate LED, trigger buzzer, display WET on LCD

05 If value ≤ threshold → DRY state, show NORMAL

WET STATE

LED ON · Buzzer SOUNDS · LCD shows WET

"If sensor value exceeds threshold, system immediately activates all alert outputs"

DRY STATE

LED OFF · Buzzer SILENT · LCD shows DRY

"System continues monitoring — no alert triggered when value is within normal range"

CODE EXPLANATION

- Initializes LCD and system components

```
#include <LiquidCrystal.h>
LiquidCrystal lcd(7, 6, 5, 4, 3, 2);

int sensorPin = A0;
int ledPin = 8;
int buzzerPin = 9;
int moistureValue;

void setup() {
  lcd.begin(16, 2);
  pinMode(ledPin, OUTPUT);
  pinMode(buzzerPin, OUTPUT);

  lcd.print("Diaper Monitor");
  delay(2000);
  lcd.clear();
}
```

- Reads moisture level using ADC

```
moistureValue = analogRead(sensorPin);

lcd.setCursor(0, 0);
lcd.print("Moisture: ");
lcd.print(moistureValue);
```

- Checks if moisture exceeds threshold
- Activates buzzer

```
if (moistureValue > 600) {
  digitalWrite(ledPin, HIGH);
  tone(buzzerPin, 2000);
  delay(200);
  noTone(buzzerPin);
  delay(200);

  lcd.setCursor(0, 1);
  lcd.print("Status: WET ");
}
```

```
else {
  digitalWrite(ledPin, LOW);
  digitalWrite(buzzerPin, LOW);

  lcd.setCursor(0, 1);
  lcd.print("Status: DRY ");
}

delay(1000);
```

- Turns off alerts when dry

SIMULATION DEMONSTRATION

POTENTIOMETER

Simulates moisture level
↕ Adjust to test

ARDUINO UNO R3

Reads ADC value
Applies threshold
Decides state

DRY

WET

BUZZER
Audio alert

LED
Visual indicator

LCD
Status display

"When potentiometer value increases → LED turns on, buzzer sounds, LCD displays diaper is WET"

CONCLUSION

Successful Detection

The system successfully detects moisture and immediately alerts caregivers in real time

Future: Wireless

Future improvements may include wireless IoT communication for remote caregiver alerts via mobile app

Improved Hygiene

Proactive monitoring improves hygiene management and patient comfort for infants and elderly patients

Future: Expanded Sensing

Temperature monitoring and multi-sensor integration can be added to enhance diagnostic capability